

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: plasmid
Species Used	Human,U937, histiocytic lymphoma		

Before the Pulse

Cell Growth Medium	RPMI 1640, 10 % Fetal Calf Serum (GIBCO/BRL, Sigma)	Growth Phase at Harvest	5 to 10 x10 (5) / ml
		Pre-pulse Incubation	Phosphate Buffered Saline (-)
Wash Solution	Phosphate Buffered Saline		

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	Room temperature	Cuvette Gap	0.4 cm
Electroporation Medium*	Phosphate Buffered Saline	Voltage	0.2 kV
Cell Density	1 x 10 (8) / ml	Field Strength	0.5 kV/cm
Volume of Cells	2 x 10 (7) / 200 µl	Capacitor	960 µF
DNA Concentration	50 µg / 200 µl	Resistor	(Pulse Controller) Ω none
DNA Resuspension Buffer	Not given		
Volume of DNA	50 µg	Time Constant	50 to 70 msec

After the Pulse	
Outgrowth Medium	RPMI 1640, 10% Fetal Calf Serum

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	1 day
Selection Method or Assay Used	G418: 700 µg / ml 3 days; then 400 µg / ml 3 days.
Electroporation Efficiency	Not assayed
Per Cent Survival	Not assayed

Name of Submitter Shumsuke Mori

Institution Address Institute for Medical Genetics
Experimental Genetics
Kuhanji 4-24-1
Kumamoto, JAPAN

Survey Number

116